

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering **ASSESSMENT TOOL 1 – Data Interpretation**

Course Code:	ITA05	Course Title:	Computer Vision
CO Assessed:	CO4 – Articulation (combining methods for evaluation) P4	Assessment:	Assessment Tool 1 – Data Interpretation
Weightage:	40% of CIA	Total Marks:	100
CO4 Statement:	<i>Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL5)</i>		
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Section / Batch:	2024 AIML	Date:	17/8/2026

Code of Conduct

I M.MANASA(192425189) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: m.manasa

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data	
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided	
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data	
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions	
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand	

1. Real-Time System Design Trade-off

Problem

A surveillance system must operate under strict latency constraints. Three techniques are considered:

Technique	Speed	Accuracy
Viola-Jones	Fast	78%
YOLO	Moderate	92%
Deep Learning	Slow	95%

Data Interpretation

Viola-Jones provides the highest speed, making it suitable for systems where immediate detection is the primary requirement. However, its accuracy of **78%** is considerably lower than the other techniques.

YOLO provides a good balance between speed and accuracy. It achieves **92% accuracy**, which is 14 percentage points higher than Viola-Jones, while still maintaining moderate processing speed.

Deep Learning provides the highest accuracy of **95%**, but its slower processing makes it less suitable when the surveillance system has strict latency requirements.

Analysis

The choice depends on the deployment condition. If the system operates on a low-power device and extremely low latency is mandatory, Viola-Jones can be considered. However, the 78% accuracy may result in many missed or incorrect detections.

YOLO offers a much better compromise. It provides high accuracy while maintaining moderate speed. In a practical surveillance system, a small delay is generally preferable to a large reduction in detection accuracy.

Although Deep Learning gives the highest accuracy, its computational requirements and slower processing may introduce unacceptable delays in real-time surveillance.

Decision

YOLO should be selected because it provides the best balance between latency and accuracy.

Justification

- Viola-Jones: Very fast but accuracy is relatively low.
- YOLO: Moderate speed with **62% accuracy**.
- Deep Learning: Highest accuracy but slow.
- For real-time surveillance, both detection quality and response time are important.

Final choice: YOLO

2. Robust Detection under Environmental Variations

Given Data

Method	Normal	Occlusion	Low Light	Average
Viola-Jones	80%	60%	55%	65.0%
YOLO	60%	75%	70%	78.3%
Deep Learning	65%	85%	80%	86.7%

Data Interpretation

All three methods perform best under normal conditions. Their performance decreases when occlusion and low-light conditions are introduced.

Viola-Jones decreases from **80% under normal conditions to 60% under occlusion and 55% under low light**. This shows that it is highly affected by environmental changes.

YOLO performs better, with 60% accuracy under normal conditions, 75% under occlusion, and 70% under low light.

Deep Learning achieves the highest performance in every condition: **65% normal, 85% occlusion, and 80% low light**.

Analysis

The average accuracy demonstrates the overall robustness:

- Viola-Jones: **65.0%**
- YOLO: **78.3%**
- Deep Learning: **86.7%**

Deep Learning has the smallest performance degradation compared with the other techniques. It therefore adapts better to difficult visual conditions.

However, Deep Learning generally requires greater computational resources and may have higher latency. YOLO provides a reasonable compromise when real-time operation is also important.

Decision

Deep Learning provides the best overall robustness.

Justification

For surveillance applications exposed to changing lighting and partial occlusion, detection reliability is more important than achieving maximum speed. Therefore, Deep Learning is the preferred choice when computational resources are available.

If the deployment device has strict computational limitations, YOLO would be a practical alternative.

Final choice for maximum robustness: Deep Learning

3. Optical Flow Reliability vs Noise Sensitivity

Given Methods

Method A

- Produces stable optical-flow vectors.
- Less affected by noise.
- May miss small or subtle movements.

Method B

- Detects fine and small movements.
- Produces noisy vectors.
- Requires additional processing to obtain stable motion information.

Comparison

Feature	Method A	Method B
Vector stability	High	Low
Small motion detection	Low	High
Noise sensitivity	Low	High
Tracking reliability	High	Moderate

Analysis

Method A is more reliable when the objective is to track large and continuous movement. Stable vectors make tracking easier and reduce false movement caused by noise.

Method B is more suitable when detecting small movements is critical. However, noisy vectors can cause unstable tracking and may produce false motion information.

The best method therefore depends on the application.

For **vehicle tracking, pedestrian tracking, or general surveillance**, Method A is preferable because stable tracking is more important.

For **gesture recognition, subtle object movement, or fine-motion analysis**, Method B may be more useful.

Decision

For a general dynamic surveillance scene, **Method A is more suitable** because reliable and stable motion tracking is usually the primary requirement.

However, Method B becomes preferable when detecting very small movements is the main objective.

Final choice for reliable dynamic-scene tracking: Method A

4. Background Modeling Strategy – GMM

Problem

A Gaussian Mixture Model (GMM)-based background subtraction system shows:

- High false positives in dynamic backgrounds.
- Missed detections in static-background situations.

Data Interpretation

GMM separates foreground objects from the background by modeling the distribution of pixel values. Dynamic backgrounds such as moving trees, waving curtains, or changing shadows can cause the model to incorrectly classify background changes as foreground.

The system may also miss objects when their appearance becomes similar to the learned background or when the model adapts incorrectly.

Scenario Comparison

Scenario	GMM Performance	Main Issue
Static background	Generally suitable	Missed objects may occur due to adaptation
Dynamic background	Reduced reliability	High false positives
Changing illumination	Reduced reliability	Pixel distribution changes
Complex scenes	Moderate	Difficult background separation

Analysis

The results indicate that **GMM alone is not sufficient for robust detection in all environments.**

A practical system should combine GMM with additional computer vision techniques.

Proposed Improvements

1. **Adaptive background modeling**
Update the background model according to environmental changes.
2. **Morphological operations**
Apply erosion and dilation to remove small noise and improve foreground regions.
3. **Object detection**
Combine GMM with YOLO to verify whether detected regions actually contain meaningful objects.
4. **Shadow removal**
Reduce false detections caused by shadows.
5. **Temporal filtering**
Track detected regions across multiple frames instead of making decisions from a single frame.

Decision

GMM should not be used alone for a robust surveillance system.

A hybrid approach such as **GMM + object detection + temporal filtering** would provide better reliability, particularly in dynamic environments.

Final recommendation: Use GMM as a preprocessing/background-subtraction stage rather than the complete detection system.

5. Motion Estimation vs Computational Cost

Given Methods

Method	Error	Computational Cost
Method A	5%	High
Method B	10%	Low

Data Interpretation

Method A provides better motion estimation because its error is only **5%**. However, it requires high computational resources.

Method B has a higher error of **10%**, but its computational requirement is low. Therefore, it can process data more efficiently.

The difference in error is:

$$10\% - 5\% = 5 \text{ percentage points}$$

Thus, Method A provides better accuracy at the cost of increased computation.

Real-Time Application

For real-time systems, computational speed and response time are important. Method B may be preferred when hardware resources are limited or when a small increase in estimation error is acceptable.

For example, a system running on an embedded device may choose Method B because its lower computational cost allows faster processing.

Offline Application

For offline applications, processing time is less critical because results do not have to be produced immediately.

Therefore, Method A is preferable because its **5% error** provides more accurate motion estimation.

Decision

Application	Preferred Method	Reason
Real-time	Method B	Low computational cost
Offline	Method A	Lower error
High-accuracy analysis	Method A	5% error
Resource-constrained system	Method B	Efficient computation

Final conclusion: Method A is better for accuracy-oriented offline processing, while Method B is more appropriate for computationally constrained real-time applications.

Technique	Speed	Accuracy
Viola-Jones	Fast	78%
YOLO	Moderate	92%
Deep Learning	Slow	95%

Overall Comparative Conclusion

The five cases demonstrate that there is no single Computer Vision technique that is optimal for every deployment condition.

Problem	Best Choice	Main Reason
Real-time surveillance	YOLO	Strong balance between speed and accuracy
Environmental robustness	Deep Learning	Highest accuracy across conditions
Dynamic-scene motion tracking	Method A	Stable motion vectors
Background modeling	Hybrid GMM approach	GMM alone produces limitations
Real-time motion estimation	Method B	Lower computational cost
Offline motion estimation	Method A	Lower error

The key trade-off is between **accuracy, robustness, computational cost, and latency**. A real-world Computer Vision system should therefore select a technique based on its deployment requirements rather than simply choosing the method with the highest accuracy.

For strict real-time surveillance, YOLO is a strong practical choice because it achieves **92% accuracy** while avoiding the slower processing associated with high-accuracy deep-learning models. For highly variable environments where accuracy is the primary requirement, Deep Learning provides the strongest performance.

Similarly, GMM is useful as a background-modeling component but should preferably be combined with object detection and temporal processing for robust deployment.

Key Insights

1. **Higher accuracy does not always mean better deployment performance.**
2. **Latency is critical in real-time surveillance systems.**
3. **Environmental variation can significantly reduce detection accuracy.**
4. **Stable optical-flow vectors are useful for reliable tracking.**
5. **GMM alone is insufficient for complex dynamic backgrounds.**
6. **Offline applications can prioritize accuracy over computational speed.**
7. **Hybrid Computer Vision systems can combine the strengths of multiple techniques.**

Final Recommendation

The appropriate Computer Vision technique should be selected according to the application's requirements:

Real-time + balanced accuracy → YOLO

Maximum robustness + computational resources → Deep Learning Stable motion tracking → Method A

Complex background subtraction → GMM + additional detection/tracking

Low-cost real-time motion estimation → Method B

High-accuracy offline motion estimation → Method A